

Technical Documentation

1. Project Overview

This is a React Native application built from scratch to serve as the frontend for an AI-powered business application. The app is designed with modularity, scalability, and maintainability in mind, leveraging modern React Native best practices.

2. Project Structure

Frontend/	
├── __tests__/	# Unit and integration tests
├── @types/	# TypeScript type definitions
├── android/	# Android native project files
├── ios/	# iOS native project files
├── src/	# Main source code
│ ├── assets/	# Static assets (fonts, icons, images)
│ ├── components/	# Reusable UI components
│ ├── constants/	# App-wide constants and mock data
│ ├── navigation/	# Navigation setup (React Navigation)
│ ├── providers/	# API and context providers
│ ├── screens/	# App screens/views
│ ├── store/	# State management (e.g., Redux, Zustand)
│ ├── theme/	# Theming, colors, fonts, global styles
│ └── utils/	# Utility functions
├── App.tsx	# App entry point
├── index.js	# Entry point for React Native
├── package.json	# Project metadata and dependencies
├── tsconfig.json	# TypeScript configuration
├── metro.config.js	# Metro bundler configuration
├── babel.config.js	# Babel configuration
├── jest.config.js	# Jest testing configuration
├── README.md	# Project overview and instructions
└── ...	# Other config and support files

3. Setup & Installation

Prerequisites

- Node.js (v16+ recommended)
- Yarn or npm
- Xcode (for iOS development)
- Android Studio (for Android development)
- React Native CLI

Installation Steps

Clone the repository

```
git clone <repo-url>
```

```
cd Frontend
```

Install dependencies

```
yarn install
```

or

```
npm install
```

iOS: Install CocoaPods dependencies

```
cd ios && pod install && cd ..
```

4. Development Workflow

- **Start Metro Bundler:**
yarn start or npx react-native start
 - **Run on Android:**
yarn android or npx react-native run-android
 - **Run on iOS:**
yarn ios or npx react-native run-ios
 - **Testing:**
yarn test or npm test
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5. Key Technologies & Dependencies

- **React Native:** Cross-platform mobile development
 - **TypeScript:** Static typing
 - **Jest:** Testing framework
 - **React Navigation:** Navigation management
 - **Other dependencies:** See package.json for a full list
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6. Core Features

- Modular, reusable UI components
 - Theming and global styles
 - Navigation with stack and tab navigators
 - State management (custom or via a library)
 - API integration (see src/providers/Api.ts)
 - Custom assets (fonts, icons, images)
 - Platform-specific configuration for iOS and Android
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7. Folder & File Descriptions

- **src/components/**: All reusable UI components (buttons, inputs, icons, etc.)
 - **src/screens/**: Main screens/views of the app (e.g., Login, Onboarding, Connect Bank)
 - **src/navigation/**: Navigation setup and route definitions
 - **src/assets/**: Fonts, icons, and images used throughout the app
 - **src/constants/**: Static data and constants
 - **src/providers/**: API and context providers
 - **src/store/**: State management logic
 - **src/theme/**: Colors, fonts, and global styles
 - **src/utils/**: Utility/helper functions
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8. Testing

- Tests are located in the `__tests__` directory.
 - Run all tests with:
`yarn test` or `npm test`
 - Uses Jest for unit and integration testing.
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9. Build & Deployment

- **Android:**
Standard React Native build process (`yarn android`)
 - **iOS:**
Standard React Native build process (`yarn ios`)
 - For production builds, follow the official React Native documentation for [Android](#) and [iOS](#).
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10. Contribution Guidelines

- Use feature branches for new work.
 - Follow the code style enforced by ESLint and Prettier.
 - Write tests for new features and bug fixes.
 - Submit pull requests for review.
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11. Troubleshooting & FAQ

- **Metro bundler issues:**
Try resetting cache: `yarn start --reset-cache`
- **iOS build issues:**
Run `cd ios && pod install`

- **Android build issues:**
Ensure Android Studio and SDKs are up to date.
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12. Additional Notes

- For any platform-specific configuration, refer to the `ios/` and `android/` directories.
 - For custom fonts and assets, ensure they are linked properly (see `react-native.config.js`).
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